(1) Solve $\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = \sin 2x$

(2) $(D^2+4D+4)y = \cos x$

(3) $(D^2-4D-5)y = e^{3x} + 4\cos 3x$

(4) $(D^2+9)y = \cos 3x$

(5) $(D^2+1)y = \sin x \sin 2x$ (6) $(2D^2-5D+3)y = \cos 3x \cos 2x$

Case III when $\phi(x) = x^m$, m being a +ve integer

Here we have to evaluate $\frac{1}{f(D)} x^m$. For that expand $\frac{1}{f(D)} = [f(D)]^{-1}$ in ascending powers of D and operate on x^m. We use the following results for expansion.

$(1-x)^{-1} = 1 + x + x^2 + x^3 + \ldots$ (1)

$(1+x)^{-1} = 1 - x + x^2 - x^3 + \ldots$ (2)

$(1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \ldots$ (3)

$(1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \ldots$ (4)

Solve the following diff. eqs.

(1) $(D^2+2D+1)y = 2x + x^2$

(2) $(D^2+D+1)y = x^2$

(3) $(D^2+25)y = x^2 + x$

Answers

(1) $(D^2+2D+1)y = 2x+x^2$

CF is got by solving $(D^2+2D+1)y = 0$

A.E is $D^2+2D+1 = 0 \Rightarrow (D+1)^2 = 0$

$\Rightarrow (D+1)(D+1) = 0 \Rightarrow D = -1, -1$

$CF = (C_1 + C_2 x)e^{-x}$

$PI = \frac{1}{D^2+2D+1}(2x+x^2) = \frac{1}{(D+1)^2}(2x+x^2)$

$= (1+D)^{-2}(2x+x^2)$

$= (1 - 2D + 3D^2 - 4D^3 + \cdots)(2x+x^2)$ using series (4)

$\therefore$ $PI = (2x+x^2) - 2D(2x+x^2) + 3D^2(2x+x^2) - \cdots$

$= (2x+x^2) - 2(2+2x) + 3(2) + 0$

$= x^2 + 2x - 4 - 4x + 6$

$= x^2 - 2x + 2$

Gen. soln. $y = CF + PI$

$= (c_1 + c_2 x)e^{-x} + x^2 - 2x + 2$

(2) $(D^2+D+1)y = x^2$

CF is got by solving $(D^2+D+1)y = 0$

A.E is $D^2+D+1 = 0$

Solving $D = \frac{-1 \pm \sqrt{1-4}}{2} = \frac{-1 \pm \sqrt{-3}}{2}$ $\quad \alpha = -\frac{1}{2}, \beta = \frac{\sqrt{3}}{2}$

$= \frac{-1 \pm i\sqrt{3}}{2}$

$CF = e^{-\frac{x}{2}}\left[c_1 \cos\frac{\sqrt{3}}{2}x + c_2 \sin\frac{\sqrt{3}}{2}x\right]$

$PI = \frac{1}{D^2+D+1}(x^2)$

$= (1+D+D^2)^{-1}(x^2)$

$= x^2 - (D+D^2)(x^2) + (D+D^2)^2(x^2) - \cdots$

$= x^2 - (2x+2) + (D^2+2D^3+D^4)(x^2) - \cdots$

$= x^2 - (2x+2) + (2+0+0) - 0$

$= x^2 - 2x - 2 + 2$

$= x^2 - 2x$

Gen. solution: $y = e^{-\frac{x}{2}}\left[c_1 \cos\frac{\sqrt{3}x}{2} + c_2 \sin\frac{\sqrt{3}x}{2}\right] + x^2 - 2x$

(3) $(D^2+25)y = x^2 + x$

CF is got by solving $(D^2+25)y = 0$

A.E. is $D^2+25 = 0 \Rightarrow D^2 = -25 \Rightarrow D = \pm 5i$

$CF = c_1 \cos 5x + c_2 \sin 5x$.

$PI = \frac{1}{D^2+25}(x^2+x) = \frac{1}{25\left(\frac{D^2}{25}+1\right)}(x^2+x)$

$= \frac{1}{25}\left(1+\frac{D^2}{25}\right)^{-1}(x^2+x)$

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$$PI = \frac{1}{25}\left(1 - \frac{D^2}{25} + \frac{D^4}{625} - \cdots\right](x^2+x)$$

$$= \frac{1}{25}\left[(x^2+x) - \frac{D^2}{25}(x^2+x) + \frac{D^4}{625}(x^2+x) + \cdots\right]$$

$$= \frac{1}{25}\left[x^2+x-\frac{2}{25}+0\right] = \frac{1}{625}(25x^2+25x-2)$$

Gen. soln: $y = c_1 \cos 5x + c_2 \sin 5x + \frac{1}{625}(25x^2+25x-2)$

Solve (1) $(D^2-2D)y = 2+3x$

(2) $(D^2-1)y = 3+2x^2$ (3) $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = x^2 + e^{4x}$

(4) $(D^2+4D+5)y = e^{2x} + \cos x + x^2$